

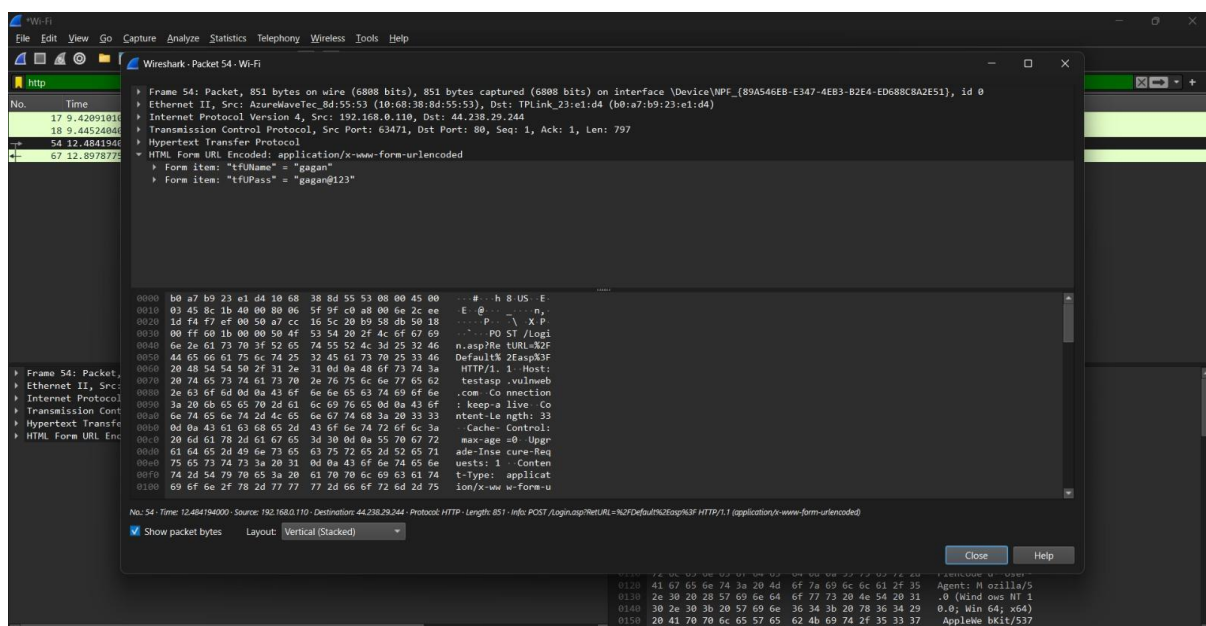
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Importance of Cryptography: Using Wireshark, investigate the transmission of dummy login credentials over HTTP and HTTPS

This experiment was to understand how login credentials are transmitted over HTTP and HTTPS. I used Wireshark to capture the network traffic and observe the difference between HTTP connection and HTTPS connection.

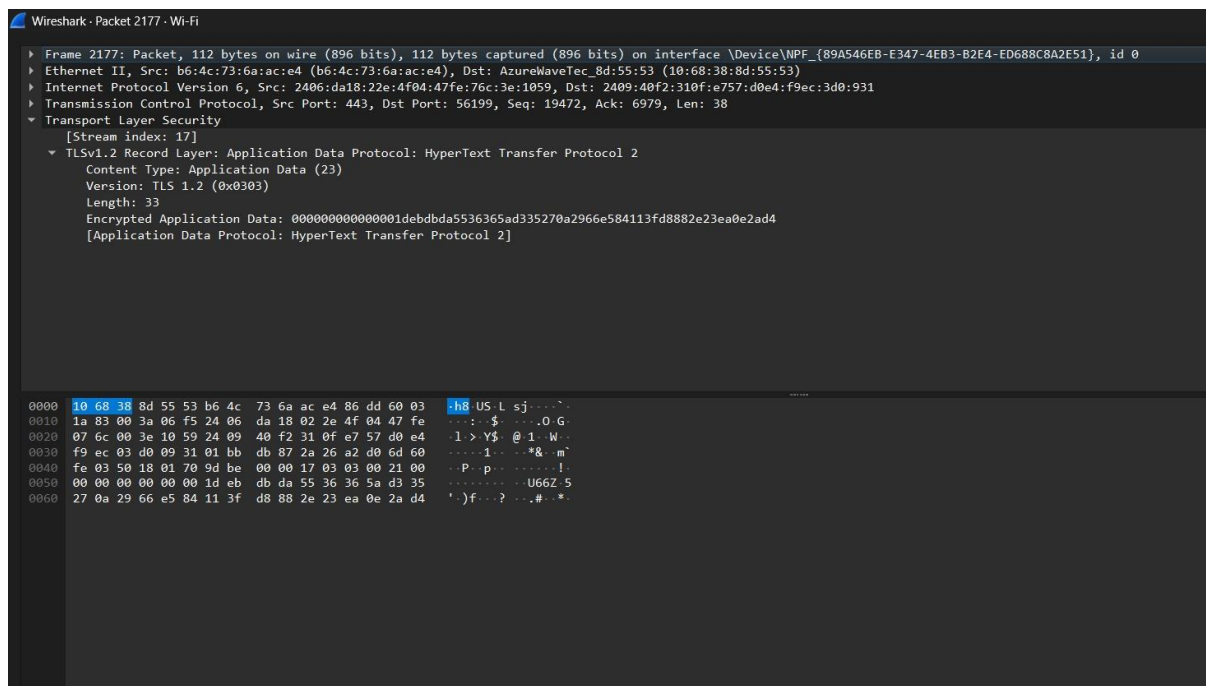
In this experiment we will open an HTTP website which contains a login page and allows us to enter Login credentials user name and password, and start wireshark to capture the packets. The same is done on HTTPS website also and the packets are captured to observe the difference.

HTTP Traffic Analysis Observations



When we log into an unsecured website (HTTP), your information is not hidden or scrambled. Anyone who intercepts the network traffic can clearly see exactly what we typed- like the username "gagan" and password "gagan@123" as normal, readable text. Because this data is completely unprotected, someone snooping on the network can easily steal your credentials. This is why it is extremely dangerous to enter passwords or personal details on websites that aren't properly secured.

HTTPS Traffic Analysis Observations



When we log into a secure website (HTTPS), our information is encrypted into an unreadable code before it leaves the device. If someone intercepts your network traffic, they won't be able to see the username or password at all.

While eavesdroppers can still see basic details - like where the information is coming from and where it is going - the actual private message is completely locked and protected. This built-in security makes it impossible for snoops to simply read your data, making secure websites essential for keeping your sensitive information safe.

Comparing HTTP and HTTPS :

- **Data Format:** HTTP sends information as readable plaintext, whereas HTTPS secures it by encrypting the data using TLS.
- **Network Port:** HTTP communicates over port 80, while HTTPS uses port 443.
- **Risk of Interception:** Because it lacks encryption, the risk of someone stealing your credentials over HTTP is high. With HTTPS, this risk is much lower.
- **Confidentiality:** HTTP offers no confidentiality for your data. HTTPS ensures confidentiality through its encryption.
- **Data Integrity:** HTTP itself cannot guarantee that your data hasn't been tampered with. HTTPS protects the integrity of your data using TLS.